

JavaScript Basics



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Agenda Today

- Introduction
- Variables & Data Types
- Conditional Statements
- Operators
- Loops
- Functions



Introduction

- JavaScript is a scripting language that is used to create interactive and dynamic website.
- JavaScript was created in **May 1995** by **Brendan Eich**, a developer at Netscape Communications Corporation.
- Interactive Website means, user can perform some action on website for example: **Button click, Submit a form, write comments, live chat.**
- Dynamic website means, the website that changes it's content or layout. Like: **Sliding effect, Ecommerce website, Quiz website.**

Features & Use-cases

1. Lightweight & Fast
2. Interpreted Language
3. Platform Independent
4. Object-Oriented
5. Event-Driven & Interactive
6. Asynchronous & Non-blocking
7. Functional & Dynamic

1. Frontend Development (Client-side)
2. Backend Development (Server-side)
3. Web Applications
4. Mobile App Development
5. Game Development
6. Desktop Applications
7. Data Visualization & Dashboards

Variables in JavaScript

- A variable is a container used to store data values.
- It gives a name to a piece of information so we can use it later in the program.
- Declaring Variables
 - ✓ var – old way, function-scoped (not recommended now).
 - ✓ let – block-scoped, can be reassigned.
 - ✓ const – block-scoped, cannot be reassigned.

```
var city = "Delhi";    // using var
let age = 25;          // using let
const pi = 3.14;       // using const
```

Data Types

1. Primitive Data types (Immutable)

- ❖ String – sequence of characters
- ❖ Number – numeric value
- ❖ Boolean – logical true/false
- ❖ Null – intentional empty value
- ❖ Undefined – variable declared but not assigned.
- ❖ Symbol(ES6) –
- ❖ BigInt(ES11/ES2020) – large integers beyond Number's safe limit

2. Non – Primitive (Reference) Data Types

- ❖ Object – collection of key-value pairs.
- ❖ Array – ordered collection of values.
- ❖ Function – block of code that can be executed.

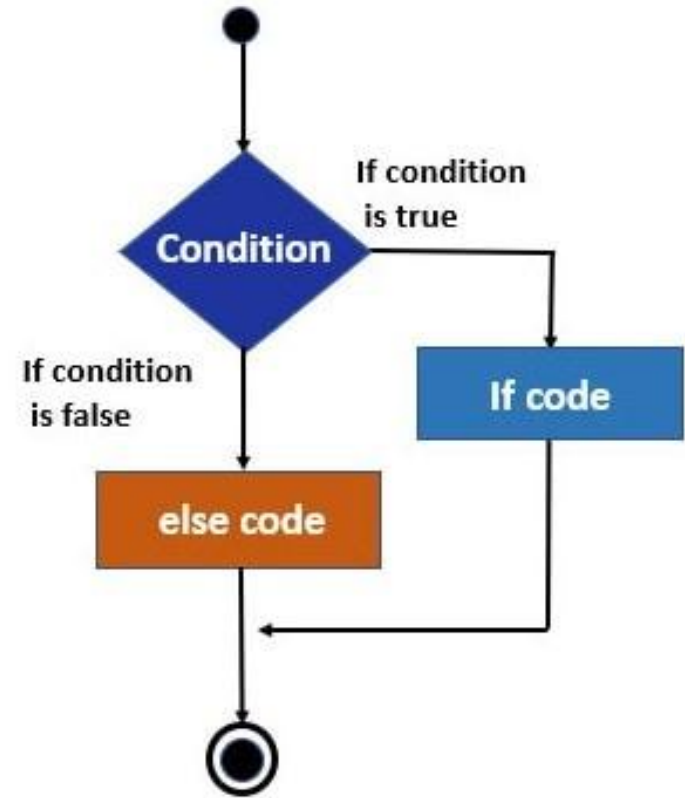
Data Types

```
let name = "Priyanshu"; // String
let age = 21;           // Number
let isStudent = true;   // Boolean
let friends = ["Saurabh", "Varun"]; // Array
let person = { name: "Varun", city: "Delhi" }; // Object
```

```
let person = { name: "Varun", city: "Delhi" }; // Object
```

Conditional Statements

- if Statement
- if...else Statement
- if...else if...else Statement
- switch Statement



Conditional Statements

```
let age = 18;  
if (age >= 18) {  
  console.log("You can vote");  
}
```

```
let age = 16;  
if (age >= 18) {  
  console.log("You can vote");  
} else {  
  console.log("You are too young");  
}
```

Conditional Statements

```
let marks = 75;
if (marks >= 90) {
  console.log("Grade A");
} else if (marks >= 75) {
  console.log("Grade B");
} else {
  console.log("Grade C");
}
```

```
let day = 3;
switch(day) {
  case 1: console.log("Monday"); break;
  case 2: console.log("Tuesday"); break;
  case 3: console.log("Wednesday"); break;
  default: console.log("Invalid day");
}
```

Operators

1. Arithmetic Operator

+ Addition- Subtraction

* Multiplication

/ Division

% Modulus (remainder)

** Exponentiation

2. Assignment Operator

= Assign

+= Add and assign

-= Subtract and assign

*= Multiply and assign

/= Divide and assign

Operators

3. Comparison Operators

`==` Equal (loose, checks value only)
`===` Strict equal (checks value + type)
`!=` Not equal
`!==` Strict not equal
`>` Greater than
`<` Less than
`>=` Greater or equal
`<=` Less or equal

4. Logical Operator

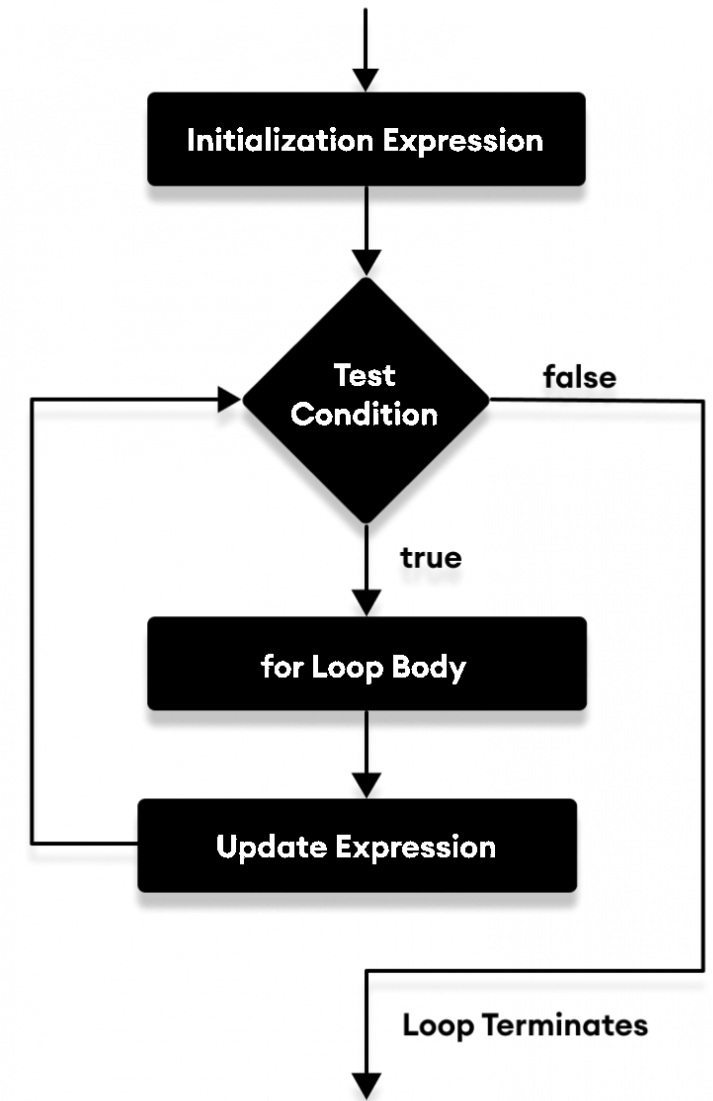
`&&` AND
`||` OR
`!` NOT

5. Ternary Operator

`condition? exp1: exp2;`

Loops

- for, while, do...while → general purpose loops.
- for...of → iterate arrays/strings.
- for...in → iterate object properties.
- Use break and continue for control.



Loops

```
for (let i = 0; i < 5; i++) {  
  console.log("Number: " + i);  
}
```

```
let i = 0;  
while (i < 5) {  
  console.log(i);  
  i++;  
}
```

```
let i = 0;  
do {  
  console.log(i);  
  i++;  
} while (i < 5);
```

Loops

```
let fruits = ["Apple", "Banana", "Mango"];  
for (let fruit of fruits) {  
  console.log(fruit);  
}
```

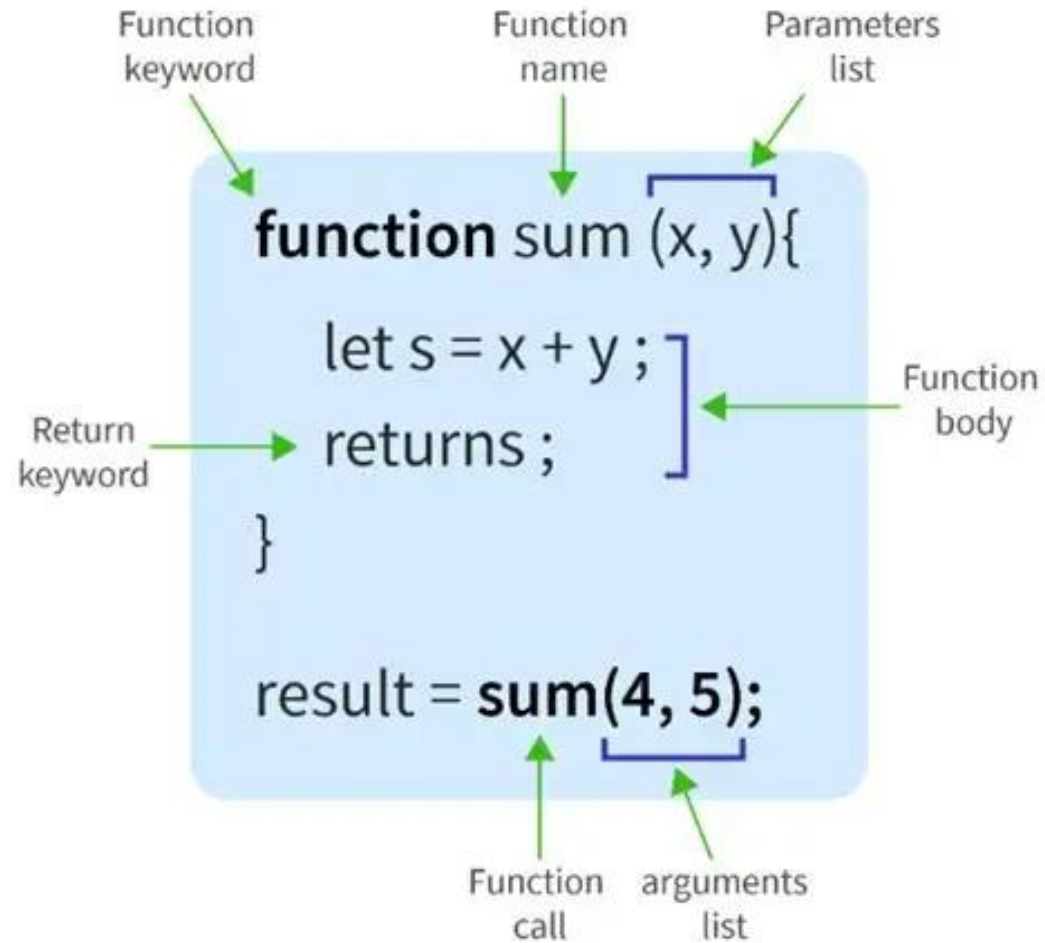
Loops

```
for (let i = 0; i < 5; i++) {  
  if (i === 2) continue; // skip 2  
  if (i === 4) break;    // stop at 4  
  console.log(i);  
}
```

Functions

- A **block of code** designed to perform a specific task.
- Can be **reused** multiple times.
- Improves **readability** and **modularity** of code.

Functions



Functions

- Defining and Calling a Function

```
// Defining a function
function greet() {
  console.log("Hello, World!");
}

// calling the function
greet();
```

```
greet();
```

```
// calling the function
```

Functions

- Function Expressions

```
function add(a, b) {  
  return a + b;  
}  
  
console.log(add(5, 3)); // 8
```

Functions

- Arrow Functions (ES6)

```
const square = (n) => n * n;  
console.log(square(6)); // 36
```

Functions

- Default Parameters

```
function greet(name = "Guest") {  
  return "Hello, " + name;  
}  
console.log(greet()); // Hello, Guest
```

Thank you

